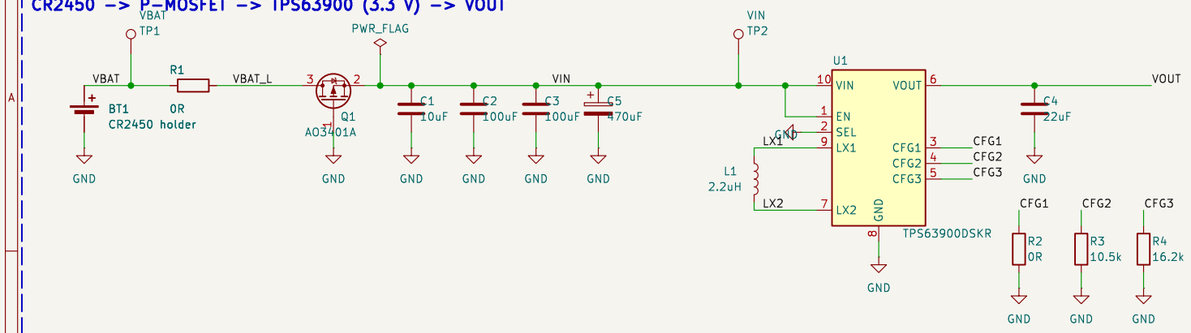
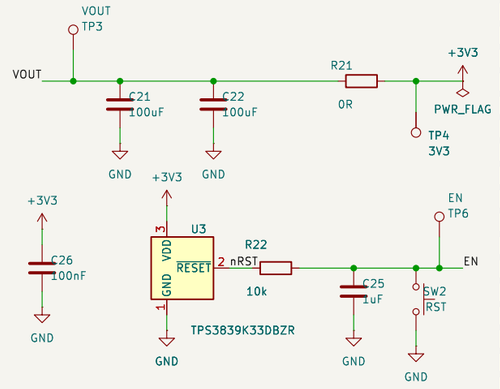
CR2450 -> P-MOSFET -> TPS63900 (3.3 V) -> VOUT

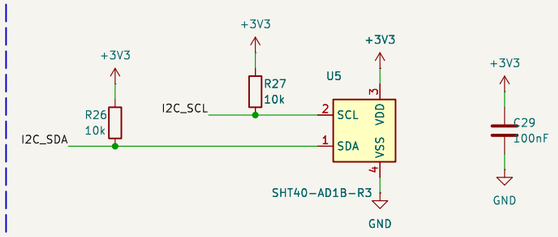


SEL = L: VO(1) = 3.3 V (RCFG3 16.2k, table 7-3). Input current limit 100 mA from RCFG1 0R / RCFG2 10.5k (table 7-2: 13.3k = 50 mA option). TI SLVSET3D, +-1 % resistors. C5: 470 uF electrolytic buffer.

3V3 RAIL, SUPERVISOR, RESET

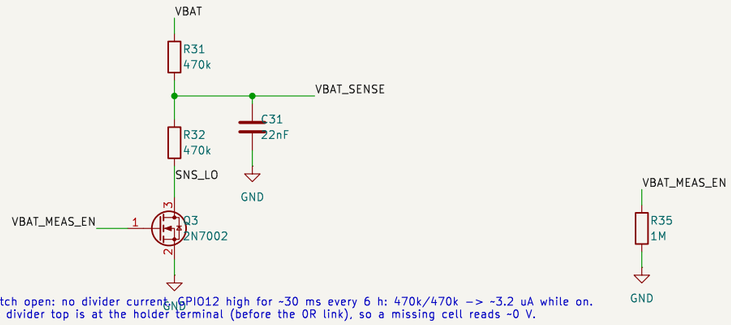


SHT40 (on a slotted tongue, away from converters and module)



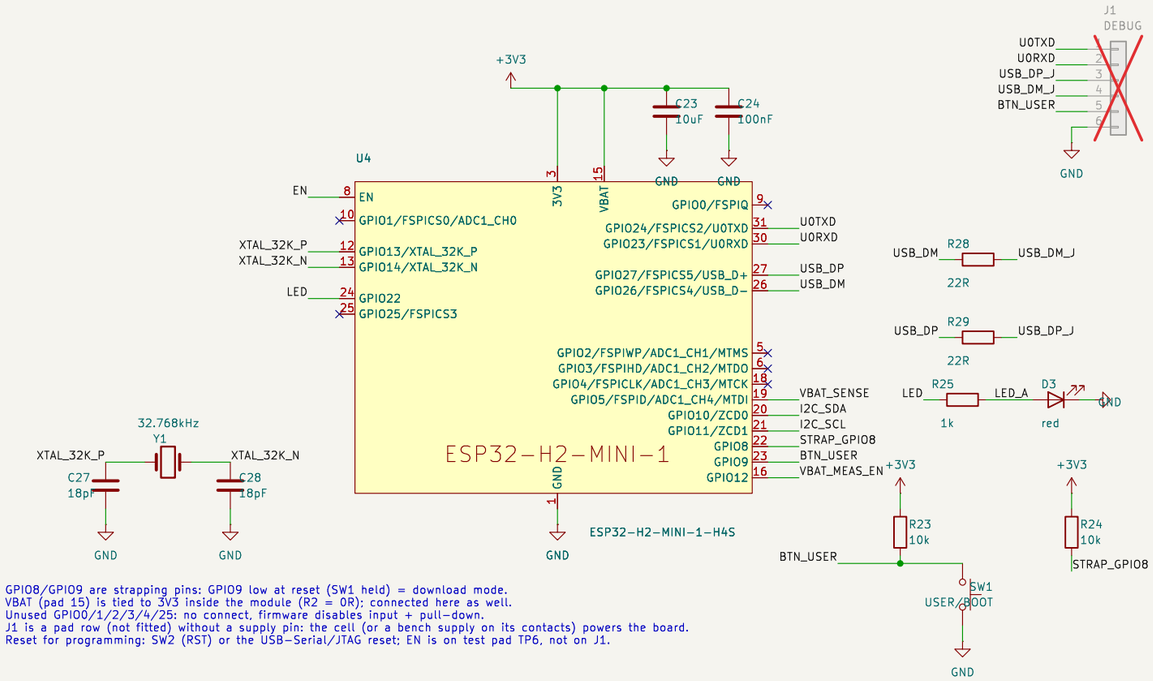
No copper under the sensor except its pads; do not solder the die pad (SHT4x datasheet).

CELL VOLTAGE SENSE (switched 470k/470k divider, ADC1_CH4 on GPIO5)



Switch open: no divider current. GPIO12 high for ~30 ms every 6 h: 470k/470k -> ~3.2 uA while on.
The divider top is at the holder terminal (before the 0R link), so a missing cell reads ~0 V.

ESP32-H2-MINI-1-H4S, 32.768 kHz CRYSTAL, USER/BOOT BUTTON, LED, DEBUG PADS



GPIO8/GPIO9 are strapping pins: GPIO9 low at reset (SW1 held) = download mode.
VBAT (pad 15) is tied to 3V3 inside the module (R2 = 0R); connected here as well.
Unused GPIO0/1/2/3/4/25: no connect, firmware disables input + pull-down.
J1 is a pad row (not fitted) without a supply pin: the cell (or a bench supply on its contacts) powers the board.
Reset for programming: SW2 (RST) or the USB-Serial/JTAG reset; EN is on test pad TP6, not on J1.

TEST PADS, MECHANICAL, NOTES



CR2450 ONLY - NOT RECHARGEABLE. No other supply can reach the rail: the cell is the only energy source.
Power-path decision and its limits: docs/decisions.md, docs/power_budget.md (ESTIMATE until measured).
Current measurement: remove R1 (cell) or R21 (load) and insert the ammeter.
U3 TPS3839K33: 2.93 V typ (2.857-2.974 V), push-pull -RESET, 200 ms delay; R22 keeps it from fighting SW2 / TP6 (EN).

Generated by hardware/scripts/gen_schematic.py from design.py - edit there
ERC/DRC reports: hardware/reports
Power path and battery life figures are ESTIMATES (docs/power_budget.md)
ESP32-H2 Matter over Thread, SHT40, 1 x CR2450, TPS63900 3.3 V

TH-SENSOR-H2

Sheet: /
File: th_sensor_h2.kicad_sch

Title: TH-SENSOR-H2 thermo-hygrometer

Size: A3	Date: 2026-10-30	**Rev: B**
KiCad E.D.A. 9.0.9		Id: 1/1